

DHARNISH GIRI

Robotics & AI Engineer · Teleoperation · ROS2 · Deep Reinforcement Learning · MPC
Dortmund, Germany · +49 151 11227434 · vgdharnishkumar@gmail.com · github.com/Dharnish08

EDUCATION

M.Sc. Automation and Robotics

Oct 2024 – Present

Technical University of Dortmund, Germany

- Relevant coursework: Deep Reinforcement Learning, Machine Learning Methods for Engineers.

B.Eng. Robotics and Automation

Aug 2019 – Apr 2023

PSG College of Technology, India

TECHNICAL PROJECTS

Franka Pick-and-Place via Hybrid RL + VLA

May 2026 – Present

Python, MuJoCo, Stable-Baselines3, SAC, SmolVLA, Gymnasium, LeRobot, Hugging Face

<https://github.com/Dharnish08/Franka-Mujoco-Teleop>

- Built a keyboard teleoperation setup in MuJoCo to record my own demonstration data for the pick task. Cartesian control of the arm runs on damped least-squares differential IK, with wrist cameras captured at every step.
- Published the dataset to the Hugging Face Hub in LeRobotDataset format, so the training data is self-collected rather than taken from an existing public set.
- Wrote a custom Gymnasium environment around the MuJoCo Menagerie Panda model. The observation space carries joint positions and velocities, gripper and box poses, and the vector between the two.
- Split the task across two controllers. A Soft Actor-Critic policy handles the approach, using a shaped reward on box-to-target and gripper-to-box distance to keep the learning signal dense.
- SmolVLA takes over for the grasp itself, fine-tuned on the collected demonstrations. RL does the reaching, the vision-language-action model does the contact-rich part.

STORM – Salient TeleOperated over Resilient Mobile Networks

Nov 2025 – Jun 2026

ROS2 Jazzy, CARLA, Python, C++, Zenoh DDS, GStreamer, Docker — Chair for Communication Networks, TU Dortmund

<https://comnets.etit.tu-dortmund.de/teaching/project-groups/storm-simulated-teleoperation-over-resilient-mobile-networks-66409/>

- Built a ROS2 Jazzy bridge between CARLA's vehicle dynamics and a Fanatec force-feedback wheel, closing the loop on human-in-the-loop driving over simulated mobile networks.
- Set up a multi-camera H.265 pipeline in GStreamer that steps bitrate and resolution down as network conditions get worse.
- Used Zenoh (rmw_zenoh_cpp) for command and telemetry traffic across two networked PCs, with every component containerised in Docker.

HUROP – Humanoid Pick-and-Place on Unitree G1

Mar 2026 – Jun 2026

Python, Isaac Lab / Isaac Sim, USD, Unitree G1, Computer Vision

<https://github.com/Dharnish08/HUROP---Unitree-G1>

- Built a perception-to-grasp pipeline for a Unitree G1 humanoid in Isaac Sim, using HSV-based colour segmentation and depth-based 3D back-projection to localise cubes from RGB-D camera output.
- Diagnosed and fixed an upstream Isaac Lab camera pose bug affecting pos_w/quat_w_ros outputs, reading transforms directly from the USD stage and converting from OpenGL to ROS camera convention.
- Validated detection accuracy against ground-truth cube layouts across 40 trials, achieving 1–2.5cm positional error, attributed primarily to centroid offset between the visible cube face and true 3D centre.

PROFESSIONAL EXPERIENCE

Student Assistant

Mar 2026 – Present

Lehrstuhl für Produktionssysteme (LPS), Ruhr-Universität Bochum

- Maintaining a PostgreSQL research database for ExoExpert, an exoskeleton development project, covering storage and retrieval of biomechanical measurement data.
- Building pipelines between measurement sessions and the analysis tooling, so control parameters can be tuned against real data instead of guesswork.

Robotics Engineer

Aug 2023 – Aug 2024

Titan Engineering and Automation Limited, India

- Programmed and commissioned ABB and Mitsubishi 6-DOF arms for production cells, writing collision-free motion programs and handling I/O-based PLC integration for the full line.
- Cut cycle time on a robotic weld cell by 15% by re-optimising trajectories and tuning dynamic parameters against live sensor feedback.
- Ran field commissioning and acceptance testing, checking motion programs against physical measurements on site.

Robotics Intern

Feb 2023 – Jul 2023

Titan Engineering and Automation Limited, India

- Wrote motion programs for 6-DOF arms on pick-and-place and palletising jobs. Hands-on with robot kinematics, TCP calibration and safety-zone setup.

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, CasADi, OpenCV, Stable-Baselines3), C++ (ROS2 nodes), MATLAB

AI & Learning: Deep reinforcement learning (SAC, PPO), vision-language-action models (SmolVLA), imitation learning, teleoperation data collection, LeRobot and Hugging Face datasets, MuJoCo, Gymnasium

Simulation: ROS2 (Jazzy / Humble), CARLA, MuJoCo, Isaac Lab / Isaac Sim, Gazebo, Nav2, MoveIt2, URDF/SRDF

Middleware: DDS, Zenoh (rmw_zenoh_cpp), ROS2 topics/services/actions, GStreamer

DevOps: Git, Docker, Ubuntu/Linux, SSH

Industrial: ABB, KUKA, Mitsubishi, Universal Robots

LANGUAGES & AVAILABILITY

English: fluent (IELTS 7.0) | German: B1, in progress | Tamil: native

Available from November 2026.